Pre-Screening Motor Imagery BCI Literacy Using Low-Sample EEG-Derived Features

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Abstract

In order to determine the practicality of assessing a person's motor imagery brain-computer interface (MI-BCI) compatibility through only low-sample EEG-derived features, we developed pre-screening models to estimate an individual's MI-BCI literacy from compact EEG summaries, enabling prospective users to gauge whether extended calibration and training are likely to be worthwhile. Rather than requiring full-length MI calibration, our approach uses a brief resting-state baseline and only the first $n = 15$ imagery trials from left-vs-right hand motor imagery runs, extracted from a small sensorimotor electrode set. We compute interpretable features capturing sensorimotor rhythm strength and separability (mu/beta band power, ERD/ERS summaries, CSP-derived class separability, SNR/variability) along with a small set of resting predictors (e.g., resting rhythm measures and spectral summaries). Ground-truth literacy is defined as each subject's cross-validated CSP-LDA decoding accuracy on the PhysioNet EEG Motor Movement/Imagery dataset Goldberger et al. (2000); Schalk et al. (2004).

Using 5-fold cross-validation, the best-performing regression model (Random Forest) achieves a moderate association with decoder-derived literacy (Pearson's $r = 0.580$; $R^2 = 0.302$ on subjects with complete labels). For an additional operational screening view, we trained a Random Forest binary classifier to separate high vs. low performers at an accuracy threshold of 0.65, achieving a LOOCV accuracy of 0.788 with stronger performance on the majority low-performing class (precision 0.833, recall 0.890) but low sensitivity to high performers (precision 0.619, recall 0.500). Overall, while low-sample EEG features may not support precise individualized performance estimation, they can reliably support coarse screening decisions, motivating rapid, data-efficient pre-screening in future MI-BCI pipelines.

Website: https://github.com/Shaheer2492/BCI-Classifer
Code: https://github.com/Shaheer2492/BCI-Classifer
https://github.com/AL2024UCSD/EEG-Project

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1 Introduction

1.1 Context

Brain–computer interfaces (BCIs) enable direct communication between neural activity and external devices by translating brain signals into actionable commands. With recent breakthroughs in neurotechnology, electroencephalography (EEG)-based BCIs have emerged as one of the most practical and widely studied approaches due to their safety, relatively low cost, and portability. In particular, motor imagery–based BCIs (MI-BCIs), which decode voluntary modulation of sensorimotor rhythms during imagined movements, have demonstrated substantial potential in assistive technologies such as neuroprosthetic control, communication systems for individuals with paralysis, and motor rehabilitation following neurological injury Nicolas-Alonso and Gomez-Gil (2012); Millán et al. (2010).

Despite these advances, a persistent limitation of MI-BCIs is the substantial inter-subject variability in performance. While most people require training to reliably use BCIs in general, a significant fraction of users, estimated to be around 20% according to Blankertz et al. (2010), fail to achieve reliable control even after training, a phenomenon commonly referred to as BCI illiteracy (in this work, BCI literacy refers to a subject's ability to achieve above-chance decoding accuracy in an MI-BCI paradigm). As a result, current BCI deployment typically requires lengthy and resource-intensive EEG recording sessions, calibration procedures, and user training phases before determining whether a given individual is compatible with a specific BCI paradigm. This trial-and-error process is costly for researchers and clinicians (up to hundreds or even thousands of dollars) and can be discouraging or emotionally taxing for prospective users Shih, Krusienski and Wolpaw (2012). This work evaluates whether low-sample and resting-state EEG features can predict MI-BCI literacy before full calibration.

1.2 Prior Research

Although prior research has identified EEG-derived correlates of MI-BCI performance, such as resting-state alpha power, sensorimotor rhythm (SMR) strength, oscillatory stability, and measures of neural complexity, there is currently no widely adopted method for preliminary assessment of MI-BCI compatibility. Most existing approaches either focus on a single predictor, require extensive task-based data, or evaluate performance only after full BCI calibration Ahn et al. (2013); Blankertz et al. (2010); Zhang et al. (2013).

One existing model, detailed in Borgheai et al. (2024), does exclusively use resting-state features but requires more extensive data collection and equipment (e.g., additional non-EEG measurements). Consequently, there remains a gap between theoretical predictors of BCI literacy and a practical, low-burden screening tool that can be applied early in the BCI pipeline. In this work, we target a *low-sample* screening setting by combining brief resting recordings with only the first 15 motor imagery trials, while keeping features interpretable and easy to compute.

1.3 Goals, Relevant Data and Features

We created BCI literacy predictors (one binary, one regression) designed to estimate an individual's compatibility with MI-based BCIs using a small set of interpretable EEG-derived features. Rather than replacing a full BCI decoder, our goal was to test the feasibility using solely lightweight, data-efficient pre-screening models that can be applied early in the pipeline.

In this study, we operationalize BCI literacy as the cross-validated decoding accuracy of a subject-specific CSP-LDA motor imagery classifier trained on the PhysioNet dataset. Our framework supports two views of "literacy". The first is a continuous prediction of expected MI decoder accuracy (regression), and the second is a binary screening decision that separates likely high vs. low performers at a predefined accuracy threshold (classification). To remain low-sample, features are derived from short resting baselines and only the first $n = 15$ imagery trials from left-vs-right motor imagery runs. The full feature set integrates complementary information capturing baseline rhythm structure (resting spectral summaries), early-trial class separability (CSP-derived separability, mu/beta bandpower), task modulation (ERD/ERS), and signal quality/consistency (SNR and trial-to-trial variability). By synthesizing these features into a single predictive model and validating it against decoder-derived MI performance, we aim to understand if it's possible to reduce the time and data required to assess MI-BCI suitability while maintaining interpretability and transparency.

2 Methods

2.1 Data Acquisition and Preprocessing

EEG data were obtained from the PhysioNet EEG Motor Movement/Imagery Dataset Goldberger et al. (2000); Schalk et al. (2004), comprising recordings from 109 healthy subjects performing motor execution and motor imagery tasks. Each subject completed runs of four paradigms: (1) left/right fist execution, (2) left/right fist imagery, (3) fists/feet execution, and (4) fists/feet imagery. Signals were recorded with BCI2000 at 160 Hz from 64 electrodes in the international 10-10 layout.

Although 64 channels are available, our analyses focus on a pre-defined sensorimotor region of interest (ROI) centered on C3, Cz, and C4, with adjacent channels FC1/FC2 and CP1/CP2 to capture nearby premotor/somatosensory activity and the spatial spread of mu/beta rhythms (see Appendix for electrode locations).

Preprocessing was performed in MNE-Python Gramfort et al. (2013). We applied common-average reference and FIR bandpass filtering (1-40 Hz). For task-based analyses, epochs were extracted from -1 to $+4$ s relative to cue onset, and epochs exceeding ± 100 μV were rejected; subjects required at least 5 valid epochs per condition. Resting-state features used baseline runs R01 (eyes open) and R02 (eyes closed). ICA artifact removal (Picard; 20 components) was used to reduce ocular artifacts using Fp1 and AF7/AF8 as EOG proxies. To improve spatial specificity, Laplacian filtering was applied for ROI channels by subtracting the mean of the four nearest neighbors; Laplacian filtering was not applied to LZC features.

Unless otherwise stated, resting features use R01/R02, while early-trial MI features use only the first $n = 15$ trials from the left-vs-right motor imagery runs (runs 4/8/12) and are computed primarily on C3/Cz/C4.

2.2 Features

All features were computed from preprocessed EEG and summarized per subject. We focus on low-burden, clear measures capturing baseline sensorimotor rhythm structure, spectral power and hemispheric specialization, and task-related modulation/complexity relevant to MI decoding.

2.2.1 Subject-Level Trait and Task-Modulation Features

Resting-state alpha over sensorimotor channels (C3/C4/Cz; primarily eyes-closed) was estimated via Welch PSD and integrated over 8-13 Hz Welch (1967). In the sensorimotor cortex this frequency range is often referred to as the mu rhythm, reflecting the same oscillatory band associated with motor imagery and movement-related desynchronization. We also computed alpha asymmetry (C3 vs. C4) and relative alpha power (alpha / total 1-40 Hz) as correlates of MI-BCI aptitude reported in prior work Wang et al. (2006) Ahn

et al. (2013).

Beta-band power was computed similarly to resting Alpha (Welch PSD; integration over band) and split into lower beta (13-20 Hz) and upper beta (20-30 Hz) to separate ERD/ERS-relevant sensorimotor activity from broader motor-state effects Schmidt et al. (2019).

Following Blankertz et al. (2010), Laplacian-filtered C3/C4 PSDs (in dB) were modeled as an aperiodic background plus oscillatory peaks. From the fit we extract SMR strength (peak height above noise floor), individual alpha frequency (IAF), alpha/beta peak amplitudes, and the aperiodic exponent, which jointly summarize rhythmic prominence and background spectral slope linked to MI-BCI performance Grandy et al. (2013).

To quantify temporal stability of alpha, we compute alpha power in short overlapping windows and summarize variability (e.g., standard deviation (SD) or coefficient of variation) over time. This captures whether subjects exhibit stronger dynamic modulation capacity in the alpha/mu range Wang et al. (2022).

Coherence between Laplacian-filtered C3 and C4 was computed using Welch-based coherence and averaged within mu (8-13 Hz) and beta sub-bands. Lower interhemispheric coupling can indicate more lateralized, decodable MI-related patterns Lee, Yoon and Lee (2020).

PSE applies Shannon entropy to the normalized PSD, yielding values near 0 for narrowband/structured spectra and near 1 for broadband/noise-like spectra Inouye et al. (1991). We compute PSE in alpha and beta bands and use task modulation $\Delta H = H_{\text{baseline}} - H_{\text{task}}$ to quantify whether MI produces more organized (lower-entropy) spectral structure.

LZC measures temporal pattern diversity by binarizing band-limited EEG (median threshold) and counting novel substrings via the LZ76 parsing algorithm, normalized by sequence length Lempel and Ziv (1976). We compute $\Delta\text{LZC} = \text{LZC}_{\text{task}} - \text{LZC}_{\text{baseline}}$ in alpha/beta bands; additionally, the real-imagined "gap" ($\Delta\text{LZC}_{\text{real}} - \Delta\text{LZC}_{\text{imag}}$) summarizes imagery fidelity, which has been linked to decodability Zhang, Roy and Jensen (2001).

TAR is defined as theta (4-8 Hz) power divided by alpha (8-13 Hz) power, computed from Welch PSD in baseline and task epochs. We use $\Delta\text{TAR} = \text{TAR}_{\text{task}} - \text{TAR}_{\text{baseline}}$ as an index of attentional/cognitive state during MI that may impact MI control Kubota et al. (2001)Ahn et al. (2013).

2.2.2 Early-Trial Decodability Features

In addition to our subject-level trait and task-modulation features, we include a second feature family designed for early prediction of MI decodability. These features are computed using only the first n trials per subject (default $n = 15$) from the left-vs-right MI runs (runs 4/8/12), using a task window of 1-4 s and a baseline window of −1-0 s. Features focus on the 3 main motor channels (C3/Cz/C4) and the mu/beta rhythms, which are central to MI decoding and ERD/ERS-based control Singh et al. (2021); Pfurtscheller and Lopes da Silva (1999).

We estimate Welch PSD on early-trial epochs and summarize mean band power in mu

(8-13 Hz) and beta (13-30 Hz) for C3, Cz, and C4. These capture sensorimotor rhythm strength commonly exploited by MI-BCI pipelines and CSP-based decoding Singh et al. (2021); Ramoser, Muller-Gerking and Pfurtscheller (2000).

We fit a CSP model on early trials and extract summary statistics describing spatial-filter usefulness: an eigenvalue/variance ratio proxy, CSP feature mean and SD, and a class-separability score (distance between class means in CSP space). These quantify how separable left/right MI appears after spatial filtering (i.e., how "decodable" the subject looks early) Ramoser, Muller-Gerking and Pfurtscheller (2000); Zhang et al. (2013).

For mu and beta bands on C3/Cz/C4, we compute relative power change from baseline to task, ERD/ERS $= \frac{P_{\text{task}} - P_{\text{base}}}{P_{\text{base}} + \epsilon}$, using Welch PSD on early trials Welch (1967). This captures event-related desynchronization/synchronization strength during MI, a canonical physiological marker used for MI control Pfurtscheller and Lopes da Silva (1999); Singh et al. (2021). It is a unitless measurement and represents the relative change in band power between task and baseline.

We quantify inter-trial consistency using (i) coefficient-of-variation summary statistics computed across trial means, (ii) average inter-trial correlation (sampled pairwise), and (iii) a stability index based on inverse variance across trials. Trial-to-trial variability has been reported to relate to individual MI performance/response differences, motivating its use as an early performance indicator Holper et al. (2012).

We compute SNR per motor channel as the ratio of mean squared amplitude during task versus baseline, and report summary statistics (mean, SD, max) across C3/Cz/C4. This approximates how strongly MI activity rises above baseline fluctuations; it is also motivated by the well-known sensitivity of CSP/covariance estimation to noise and outliers Zhang et al. (2013); Azab et al. (2019).

2.3 Dataset Construction and Statistical Analysis

All extracted features were aggregated to one vector per subject and merged by subject ID into a single design matrix. The merged feature set combines early-trial MI decodability features computed from only the first $n = 15$ imagery trials, additional engineered rhythm-strength/ERD summaries from our EDA, and a small subset of resting-state predictors computed from R01/R02. This produces a fixed-length feature vector per subject (38 total merged features).

To reduce redundancy and control false discoveries, we performed supervised feature selection using rank-based association (Spearman), permutation-based significance testing, and Benjamini-Hochberg FDR correction Benjamini and Hochberg (1995). The final regression model was trained on the selected feature subset (12 predictors). When a subject was missing a feature group, we used feature-wise median imputation to preserve cohort size. Feature selection was performed on the full feature matrix prior to model training to identify the most informative predictors; however, future work should incorporate nested cross-validation so that feature selection occurs strictly within training folds to avoid potential information leakage.

2.4 Ground-Truth MI Decoder (CSP-LDA)

To establish ground-truth MI performance, we implemented a subject-specific CSP-LDA decoder following the MetaBCI framework Mei et al. (2024); Blankertz et al. (2010). EEG was filtered to the mu band (8-13 Hz) and cropped to a post-cue window (0.5-4.0 s). For each subject, we fit CSP and retained the top $k = 4$ and bottom $k = 4$ spatial filters (8 components total); log-variance features from CSP-projected signals were classified using LDA with shrinkage regularization. Decoder accuracy was evaluated with stratified 5-fold cross-validation within subject to produce a robust subject-level "BCI literacy" score used as the downstream prediction target.

2.5 BCI Literacy Screening Models

We trained supervised models to predict the decoder-derived literacy score from the merged feature vectors. Features were standardized prior to model fitting. We compared four regression models (Random Forest, Gradient Boosting, RBF-SVR, and Ridge regression) Breiman (2001); Friedman (2001); Chen and Guestrin (2016); Ke et al. (2017) using 5-fold cross-validation, selecting the best model by Pearson's r and mean R^2 while also reporting RMSE/MAE and correlation with ground-truth accuracy.

For an operational screening decision, we additionally trained a Random Forest binary classifier that labels subjects as high-performing if decoder accuracy ≥ 0.65 and low-performing otherwise, as previous studies determined that to be around the threshold between BCI-literate and BCI-illiterate people Ahn et al. (2013). This classifier was evaluated using leave-one-out cross-validation (LOOCV), and we report accuracy and class-wise precision/recall/F1.

3 Visualizations

To help us and others gain a better understanding of our features, we visualize "extreme phenotype" subjects (highest vs. lowest feature values) to make abstract features more interpretable and to connect feature values to qualitative EEG patterns. We also created tables showing feature significance and model performance.

3.1 Fig 1. Normalized PSD curves (PSE extreme)

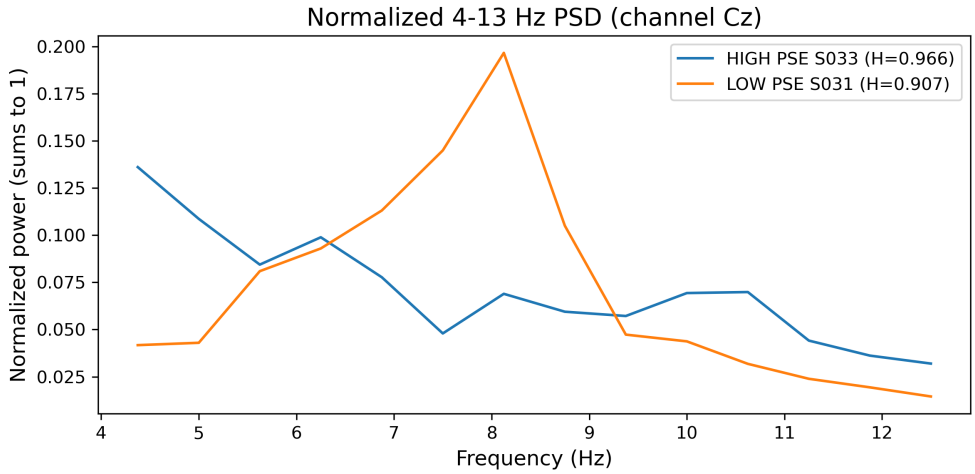

Figure 1 compares the average power distribution across frequency bands for high vs. low motor-imagery PSE in the Theta-Alpha frequency range. High-entropy subjects show power being distributed across frequency bands, while low-entropy subjects have power concentrated in a select few frequencies (in this case our low-entropy subject has power heavily concentrated at and around 8 Hz) (Inouye et al. (1991)). This supports PSE as a proxy for the degree of organized sensorimotor rhythm engagement during MI.

3.2 Fig 2. Channel-wise TAR comparison (TAR extreme)

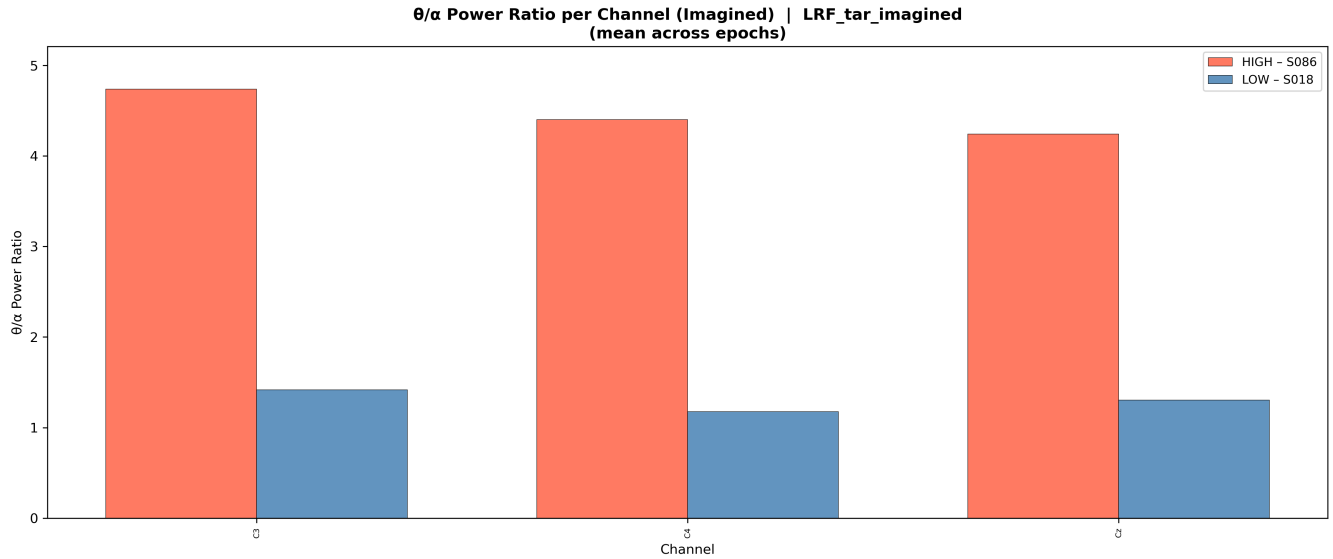

Figure 2 compares TAR across C3/Cz/C4 for high- vs. low-TAR subjects during imagery. Higher TAR indicates relatively stronger theta vs. alpha activity (often associated with increased cognitive load or reduced readiness), while lower TAR suggests a more favorable attentional state for consistent MI modulation (Kubota et al. (2001); Klimesch (1999)).

The channel-wise view checks whether effects are global or localized over sensorimotor cortex.

3.3 Fig 3. Recurrence plots (LZC extreme)

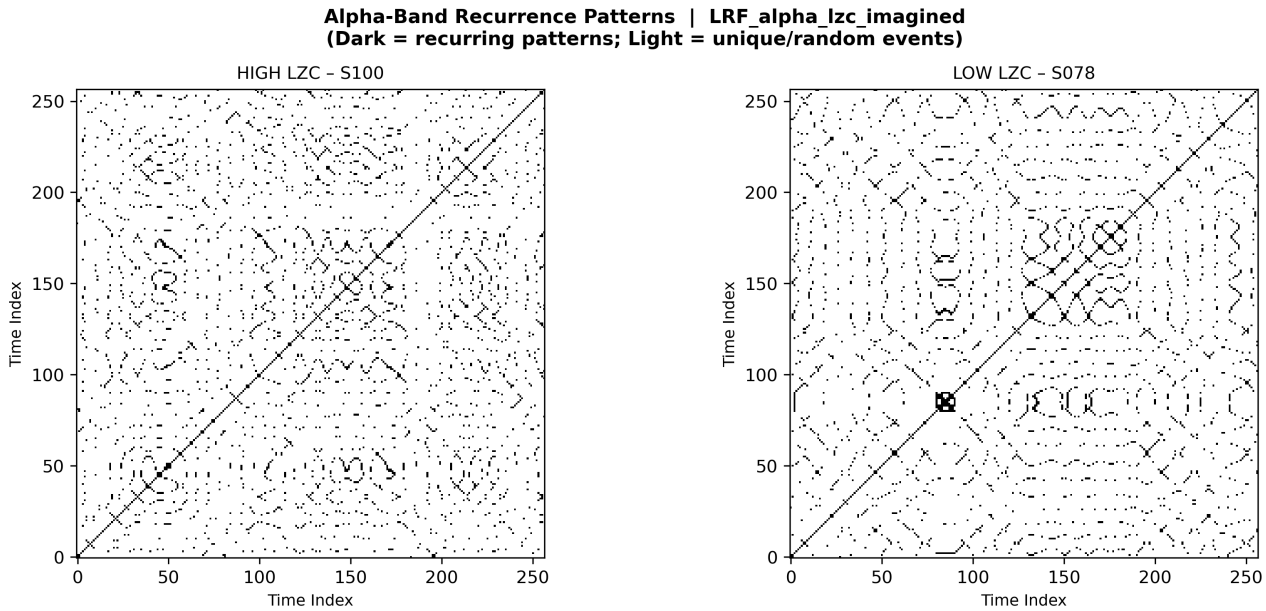

Figure 3 shows recurrence plots for alpha-band signals in high- vs. low-LZC subjects. Band-limited alpha signals will almost always show some structure because they're oscillatory. However, the high LZC corresponds to more diffuse recurrence structures (fewer repeated states, more speckles), whereas lower LZC shows more repeated motifs/diagonal structure consistent with more clean dynamics (Lempel and Ziv (1976); Marwan et al. (2007)). The contrast provides an intuitive depiction of temporal pattern diversity underlying LZC. Note that in the final models we did not incorporate LZC as it had negligible significance compared to other features.

3.4 Fig 4: Model performance comparison (regression)

Model	**RMSE**	**MAE**	**R^2**	**Pearson r**	**Spearman ρ**
Random Forest	0.125 ± 0.020	0.099 ± 0.015	0.302 ± 0.097	0.580	0.474
Gradient Boosting	0.134 ± 0.019	0.107 ± 0.017	0.169 ± 0.201	0.524	0.428
SVM	0.131 ± 0.028	0.105 ± 0.021	0.246 ± 0.112	0.492	0.413
Ridge Regression	0.128 ± 0.025	0.101 ± 0.017	0.269 ± 0.131	0.530	0.428

In Figure 4, to compare candidate regression approaches for predicting decoder-derived literacy, we evaluated Random Forest, Gradient Boosting, RBF-SVR, and Ridge regression using 5-fold cross-validation Breiman (2001); Friedman (2001); Chen and Guestrin (2016); Ke et al. (2017). Random Forest performed best overall, achieving the lowest error

(RMSE/MAE) and the highest R^2, Pearson correlation, and Spearman correlation, and was therefore selected as our primary regression model. Given its superior regression performance and ability to model nonlinear feature interactions, we also selected Random Forest for the binary classification task.

3.5 Fig 5: Spearman Feature Selection Results (FDR $\alpha = 0.05$, significant features only)

Feature	ρ	p_{raw}	p_{FDR}
band_power_beta_C3	−0.264	0.008	0.036
band_power_beta_C4	−0.258	0.011	0.038
csp_class_separability	0.454	0.000	0.002
erdrs_mu_C3	−0.365	0.000	0.002
erdrs_mu_Cz	−0.257	0.011	0.038
snr_mean	−0.247	0.013	0.041
rpl	0.314	0.001	0.008
smr_strength	0.399	0.000	0.002
mu_erd_imagined_C3	−0.251	0.011	0.038
resting_rpl_alpha	0.293	0.004	0.023
resting_tar	−0.239	0.016	0.046
resting_pse_avg	−0.284	0.006	0.030

In Figure 5, to characterize which low-sample features contribute most to prediction, we report a supervised feature selection table using Spearman correlation with permutation-based p-values and Benjamini-Hochberg FDR correction and Random Forest feature importances for the final selected subset. Across both views, the strongest predictors include CSP class separability, SMR strength, rhythm-length/power summaries (RPL/beta power), ERD/ERS measures in the mu band, and a small set of resting predictors (resting TAR, resting PSE, resting RPL_α), supporting the interpretation that both separability and baseline rhythm inform BCI literacy screening Blankertz et al. (2010); Zhang et al. (2013); Ahn et al. (2013); Singh et al. (2021).

3.6 Fig 6: RF Classifier Performance Table

Class	n	Precision	Recall	F1
LOW	73	0.833	0.890	0.861
HIGH	26	0.619	0.500	0.553

Overall Metric	Value
LOOCV Accuracy	78.8%

Fig 6 summarizes the final Random Forest binary screening model, which classifies subjects as LOW vs. HIGH performers using a decoder-accuracy threshold of 0.65. Performance was estimated using leave-one-out cross-validation (LOOCV), and we report per-class precision, recall, and F1 alongside overall LOOCV accuracy Stone (1974). Class-wise metrics are computed from the aggregated LOOCV predictions across all subjects, reflecting how reliably the model flags likely low performers versus identifies high performers.

4 Results

We constructed a merged subject-level feature representation combining early-trial MI decodability features computed from the first $n = 15$ imagery trials (runs 4/8/12; channels C3/Cz/C4; 7-30 Hz), selected EDA-derived rhythm features (e.g., SMR/RPL/ERD summaries), and statistically validated resting-state predictors (resting RPL_α, resting TAR, resting PSE). The merged dataset contains 38 features for all 109 subjects.

We trained and compared four regressors (Random Forest, Gradient Boosting, RBF-SVR, Ridge) using 5-fold cross-validation on subjects with both features and ground-truth decoder labels ($N = 99$). Figure 4 summarizes RMSE, MAE, R^2, Pearson correlation, and Spearman correlation for each model. Random Forest performed best overall (RMSE 0.125± 0.020, MAE 0.099 ± 0.015, $R^2 = 0.302 \pm 0.097$, Pearson's $r = 0.580$, Spearman $\rho = 0.474$) and was selected as our primary regression model Breiman (2001).

Using Spearman correlation with permutation-based p-values (5000 permutations) and Benjamini-Hochberg FDR correction ($\alpha = 0.05$), we retained 12 predictors for training (Figure 5). The largest Random Forest importances were assigned to CSP class separability and SMR strength, followed by resting spectral summaries (resting PSE, resting TAR), RPL, ERD/ERS (mu band), and beta power features.

For a practical screening decision, we trained a Random Forest classifier using an accuracy threshold of 0.65 to define high vs. low performers. Under LOOCV, the classifier achieved accuracy 0.788 with confusion matrix $\begin{bmatrix} 65 & 8 \\ 13 & 13 \end{bmatrix}$ (rows = true LOW/HIGH; columns = predicted LOW/HIGH). Performance was stronger for the majority LOW class (precision 0.833, recall 0.890) than for the HIGH class (precision 0.619, recall 0.500), indicating the model is more reliable at identifying likely low performers than detecting high performers Kohavi et al. (1995). This sensitivity to low performers is intentional and further encouraged by a 2:1 misclassification penalty for low performers, and is consistent with our goal of flagging illiterate users for the purpose of informing these individuals of their low performance potential. However, a class imbalance of 73 low performers and 26 high performers further biases our classifier towards these results.

5 Discussion

Across models, the combined low-sample feature representation yielded consistent predictive signal for decoder-derived MI literacy. Random Forest regression outperformed Gradient Boosting, RBF-SVR, and Ridge on average, suggesting the mapping from compact EEG summaries to literacy is moderately non-linear and benefits from interaction modeling.

The classifier's reduced sensitivity to high performers likely reflects both class imbalance and the conservative nature of the screening objective. The binary classifier provides an operational framing of screening: the model achieves strong recall for the majority low-performing class, meaning it can effectively flag individuals who are unlikely to achieve reliable control under the tested MI paradigm. However, sensitivity for high performers is low, so the classifier is best interpreted as a conservative filter rather than a definitive "pass" decision. In practice, this supports a two-stage workflow: use low-sample screening to identify likely low performers early, then reserve longer calibration or alternative paradigms for borderline/high-confidence candidates Shih, Krusienski and Wolpaw (2012). Future work could experiment with different class weightings or a sample containing a more balanced distribution of low and high performers.

Feature importance patterns are consistent with established MI-BCI mechanisms: CSP class separability and SMR strength dominate, with additional contributions from ERD/ERS and beta power measures (early-trial decodability) and a small set of resting predictors (resting TAR/PSE/RPL_α). These results suggest that both baseline rhythm structure and early-trial separability carry complementary information about literacy.

5.1 Potential Improvements, Future Work

A key methodological requirement for final reporting is that scaling and supervised feature selection occur strictly within each cross-validation fold to avoid information leakage and inflated estimates. We also restricted filtering to the mu band (8–13 Hz) as it represents the primary oscillatory correlate of sensorimotor activity during MI Pfurtscheller and Lopes da Silva (1999). While standard CSP-LDA pipelines commonly include the beta band (8–30 Hz) Zhang et al. (2013), which may contribute additional discriminative information, we prioritized mu-band activity to minimize spectral noise from non-motor sources. Future work should evaluate whether extending to 8–30 Hz improves ground-truth decoder sensitivity. Additional improvements include reporting nested or pipeline-based cross-validation results and validating generalization on additional datasets and populations. Cawley and Talbot (2010); Varma and Simon (2006).

6 Conclusion

We developed a low-sample pre-screening pipeline for MI-BCI literacy that combines a brief resting baseline with only the first $n = 15$ motor imagery trials to compute compact, in-

terpretable EEG-derived features. The best regression model (Random Forest) achieved Pearson's $r = 0.580$ and $R^2 = 0.302$ against decoder-derived ground truth on subjects with complete labels ($N = 99$). For practical screening, a Random Forest classifier at a 0.65 accuracy threshold achieved LOOCV accuracy 0.788, with high reliability in identifying low performers and moderate sensitivity to high performers. Overall, these results support the feasibility of rapid, data-efficient pre-screening to reduce wasted calibration time, while motivating future work on leakage-free evaluation, improved high-performer detection, and external validation.

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Appendices

A.1 Supplementary Graphics A1

A.1 Supplementary Graphics

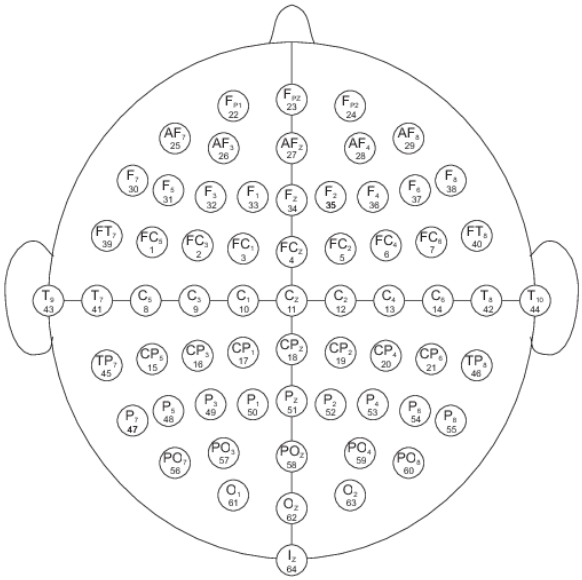

Image showing the distribution of electrodes in the EEG headset used to create the PhysioNet EEG Motor Movement/Imagery Dataset. Sensorimotor ROI electrodes used in this study (C3, Cz, C4, FC1, FC2, CP1, CP2) are located near/around the center of the scalp.